

Task 4.1: Scenario Response

Reviewer's question = "Why did you choose this specific PR over the others? What made it comprehensible to you, and what challenges do you anticipate in implementing it?"

I chose PR #196 (separate socket groups) because it was more interesting from a system design and networking perspective. Compared to PR #193 this PR works on the core connection management part of the library which made it more challenging and useful to understand.

What made the PR easier to understand was that the problem was very clearly explained. The issue was that one socket was being shared for all requests so long fetch requests could delay important heartbeat messages. This could lead to unnecessary consumer rebalancing in Kafka. The solution of creating separate socket groups for different request types felt logical and easy to follow.

I was also able to understand the implementation idea because the changes mainly involved modifying the connection mapping from a single `node_id` key to a `(node_id, group)` structure. Since I already understand basic async networking and connection handling in Python, the flow of the PR made sense to me.

One challenge I expect is handling connection failures correctly. If one socket group fails the other group should continue working normally. A wrong implementation could accidentally close all connections for the same broker node.

Another challenge is making sure the COORDINATION socket is not created for standalone consumers that do not use a `group_id`. Missing this condition could create unnecessary connections.

To solve these problems I would first update the connection structure and then test each feature step by step. I would also use integration tests to check whether heartbeat requests still work properly when long fetch requests are running.

I declare that all written content in this assessment is my own work, created without the use of AI language models or automated writing tools. All technical analysis and documentation reflects my personal understanding and has been written in my own words.